



Stony Brook University

DATA ANALYSIS

AMS 572

Chapter 7: Overview of Power and sample size calculation for one population mean & Inference on one population variance

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Power calculation for one population mean
 μ

Type I error, Type II error and Power

		Truth	
		H_0	H_a
Decision	Do not reject		Type II error
	Reject	Type I error	Power

$$\alpha = P(\text{Type I error}) = P(\text{Reject } H_0 | H_0)$$

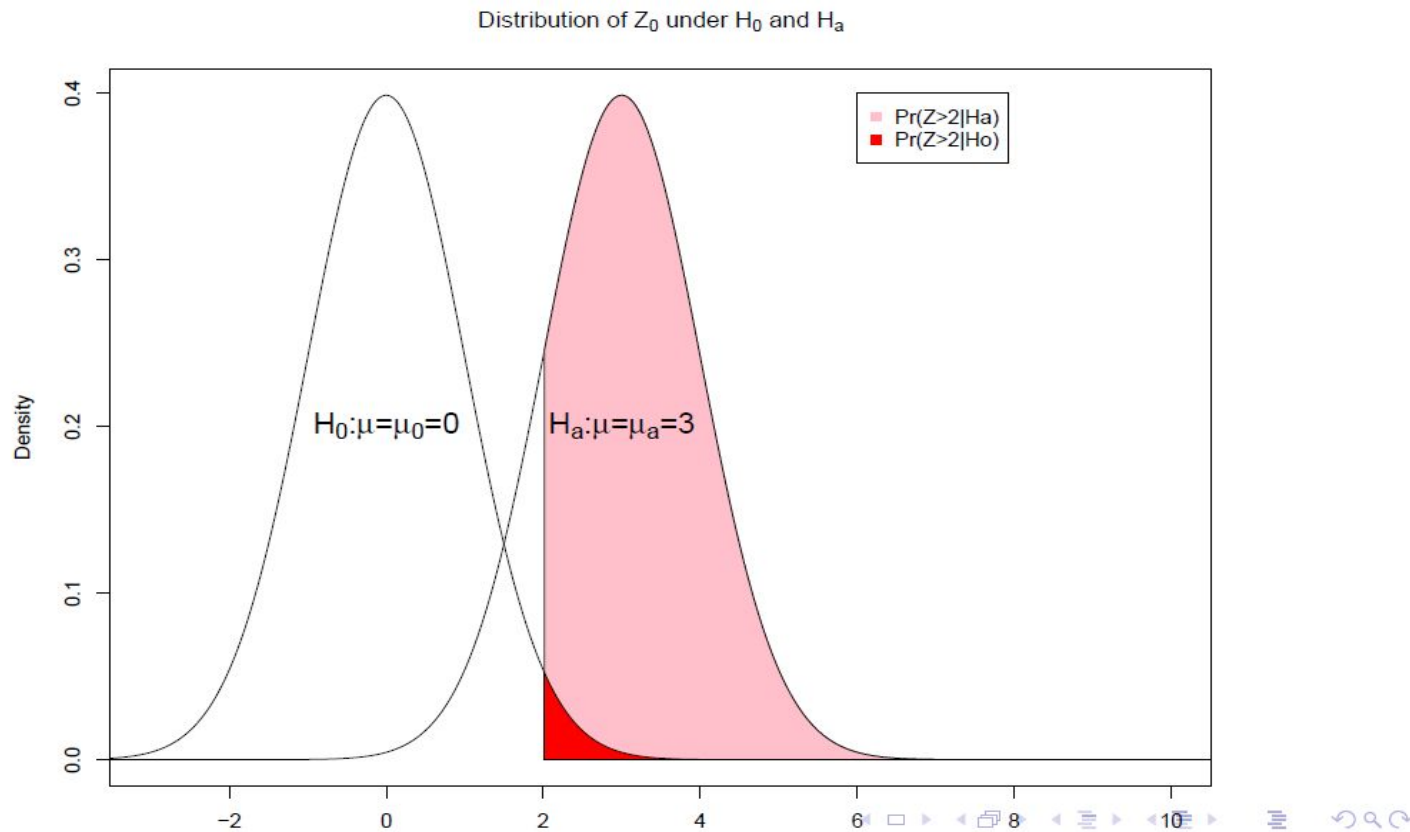
$$\beta = P(\text{Type II error}) = P(\text{Fail to reject } H_0 | H_a)$$

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 | H_a)$$

Case 1: Power calculation for normal population, σ^2 is known

- ▶ Suppose we are testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a > \mu_0$
- ▶ Test statistic : $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ under H_0 .
- ▶ At the significance level α , we reject H_0 if $Z_0 \geq z_\alpha$
- ▶ Power = $1 - \beta = P(\text{reject } H_0 | H_a) = P(Z_0 \geq z_\alpha | \mu = \mu_a)$
- ▶ What is the distribution of Z_0 when $\mu = \mu_a$?

Case 1: Power calculation for normal population, σ^2 is known



Case 1: Power calculation for normal population, σ^2 is known

- ▶ For hypothesis test $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a < \mu_0$
- ▶ Test statistic : $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ under H_0 .
- ▶ At the significance level α , we reject H_0 if $Z_0 \leq -z_\alpha$
- ▶ Power = $P(\text{reject } H_0 | H_a) = P(Z_0 \leq -z_\alpha | \mu = \mu_a)$

$$\begin{aligned} &= P\left(\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \leq -z_\alpha | \mu = \mu_a\right) \\ &= P\left(\frac{\bar{X} - \mu_a}{\sigma/\sqrt{n}} + \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} \leq -z_\alpha | \mu = \mu_a\right) \\ &= P\left(Z \leq -\frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} - z_\alpha | \mu = \mu_a\right), Z \sim N(0, 1) \end{aligned}$$

Case 1: Power calculation for normal population, σ^2 is known

- ▶ For hypothesis test $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a \neq \mu_0$
- ▶ Test statistic : $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ under H_0 .
- ▶ At the significance level α , we reject H_0 if $|Z_0| \geq z_{\alpha/2}$
- ▶ Power = $P(\text{reject } H_0 | H_a) = P(|Z_0| \geq z_{\alpha/2} | \mu = \mu_a)$

$$= P(Z_0 \geq z_{\alpha/2} | \mu = \mu_a) + P(Z_0 \leq -z_{\alpha/2} | \mu = \mu_a)$$

$$= P\left(\frac{\bar{X} - \mu_a}{\sigma/\sqrt{n}} + \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} \geq z_{\alpha/2} | \mu = \mu_a\right)$$

$$+ P\left(\frac{\bar{X} - \mu_a}{\sigma/\sqrt{n}} + \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} \leq -z_{\alpha/2} | \mu = \mu_a\right)$$

$$= P\left(Z \geq z_{\alpha/2} - \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} | \mu = \mu_a\right)$$

$$+ P\left(Z \leq -z_{\alpha/2} - \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} | \mu = \mu_a\right)$$

.....

Case 2: Power calculation for normal population, σ^2 is unknown, small sample

- ▶ Suppose we are testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a > \mu_0$
- ▶ Test statistic : $T_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$ under H_0 .
- ▶ At the significance level α , we reject H_0 if $T_0 \geq t_{n-1, \alpha}$
- ▶ Power = $1 - \beta = P(\text{reject } H_0 | H_a) = P(T_0 \geq t_{n-1, \alpha} | \mu = \mu_a)$
- ▶ What is the distribution of T_0 when $\mu = \mu_a$?

Case 2: Power calculation for normal population, σ^2 is unknown, small sample

Note that if $U \sim N(\mu, 1)$ and $V \sim \chi_k^2$ and U independent of V , then $\frac{U}{\sqrt{V/k}}$ is a non-central t distribution with k degrees of freedom and non-centrality parameter μ

Power = $1 - \beta = P(\text{reject } H_0 | H_a) = P(T_0 \geq t_{n-1, \alpha} | \mu = \mu_a)$
which can be computed using non-central t distribution.

- Reasonably good approximation to the power of the t-test can be obtained from the z-test. If z-test is used, there is a slight over estimation of the power.

Case 2: Power calculation for normal population, σ^2 is unknown, small sample

- ▶ Note: Shapiro-Wilk test can be used to determine whether the population is normal
 - ▶ Power derivation for testing
 - ▶ $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a < \mu_0$
 - ▶ $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a \neq \mu_0$
- is similar to Case 1 and thus omitted.

Case 3: Power calculation for any population, large sample

- ▶ Suppose we are testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a > \mu_0$
- ▶ Test statistic : $Z_0 = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1)$ under H_0 .
- ▶ At the significance level α , we reject H_0 if $Z_0 \geq z_\alpha$
- ▶ Power of the test = $P(\text{reject } H_0 | H_a) = 1 - \beta$ is similar to Case 1 and thus omitted.

Case 3: Power calculation for any population, large sample

- ▶ Power derivation for testing
 - ▶ $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a < \mu_0$
 - ▶ $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a \neq \mu_0$
- is similar to Case 1 and thus omitted.

Sample size determination for one
population mean μ

Introduction

- ▶ Choosing an appropriate sample size is not just a study design issue, it is an ethical issue
- ▶ For a (new) study to be ethical, it must be designed to have sufficient power to detect meaningful differences
- ▶ There are ethical issues even if using already-collected data
 - wasting resources if the sample size is too small
- ▶ Power and sample size are mathematically related
- ▶ In some situations we can calculate sample sizes explicitly
- ▶ In complicated situations, one may need to use simulation to determine the sample size

Introduction

To estimate sample size we need to specify:

- ▶ The study design
- ▶ The significance level
- ▶ The test statistic and its distribution
- ▶ The null hypothesis
- ▶ The value μ_a that we want to be able to detect
- ▶ The desired power to detect this μ_a
- ▶ More complex models may require specifying other parameters, such as covariances (for measures taken at multiple time-points or if adjusting for confounders)

These need to be specified when designing the study

Introduction

How do we decide which values to use?

- ▶ Some are reasonably standard, e.g. $\alpha = 0.05$
- ▶ Obtain estimates from pilot studies or studies done elsewhere, e.g. μ_a , variances, covariances
- ▶ For example, in clinical studies, μ_a may be what is regarded as the smallest clinically meaningful effect
- ▶ We often calculate sample size for a few representative values of what the underlying parameters might be
- ▶ We often calculate the sample size for two or more choices of the study power (typically 0.8 and 0.9)
- ▶ We often choose a few sample sizes and calculate the associated power

Case 1a: Sample size determination for normal population, σ^2 is known, for a given power

- ▶ Suppose we are testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a > \mu_0$

Case 1a: Sample size determination for normal population, σ^2 is known, for a given power

► Suppose we are testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a < \mu_0$

►

$$1 - \beta = P(Z \leq -\frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} - z_\alpha | \mu = \mu_a), Z \sim N(0, 1)$$

$$z_\beta = -\frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} - z_\alpha = \frac{\mu_0 - \mu_a}{\sigma/\sqrt{n}} - z_\alpha$$

$$\sqrt{n} = \frac{(z_\alpha + z_\beta)\sigma}{\mu_0 - \mu_a}$$

$$\therefore n = \frac{(z_\alpha + z_\beta)^2 \sigma^2}{(\mu_0 - \mu_a)^2}$$

Case 1a: Sample size determination for normal population, σ^2 is known, for a given power

- ▶ Suppose we are testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a \neq \mu_0$



$$1 - \beta = P(Z \geq z_{\alpha/2} - \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} | \mu = \mu_a) \\ + P(Z \leq -z_{\alpha/2} - \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} | \mu = \mu_a)$$

Assume $\mu_a > \mu_0$. Then,

$P(Z \leq -z_{\alpha/2} - \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} | \mu = \mu_a) \rightarrow 0$. So, we can neglect it.

Case 1a: Sample size determination for normal population, σ^2 is known, for a given power

$$\begin{aligned}1 - \beta &\approx P(Z \geq z_{\alpha/2} - \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} | \mu = \mu_a) \\-z_{\beta} &\approx z_{\alpha/2} - \frac{\mu_a - \mu_0}{\sigma/\sqrt{n}} \\n &\approx \frac{(z_{\alpha/2} + z_{\beta})^2 \sigma^2}{(\mu_a - \mu_0)^2}\end{aligned}$$

Sample size calculation for the two sided test differs from that for the one-sided tests in two aspects:

1. α is replaced by $\alpha/2$
2. It is an approximate formula (a more conservative estimate)

Case 1b: Sample size determination for normal population, σ^2 is known, for a given CI length

$$\text{P.Q. } Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

$$100(1 - \alpha)\% \text{ CI for } \mu : \bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$



Case 1c: Sample size determination for normal population, σ^2 is known, for a given margin of error E

$$P(|\bar{X} - \mu| \leq E) = 1 - \alpha$$

Case 2: Sample size determination for normal population, σ^2 is unknown

- ▶ Exact sample size calculation is based on non central t distribution.
- ▶ Reasonably good approximation to the required sample size can be obtained from the z-test. If z-test is used, there is a slight under estimation of the sample size.

Example 1: Jerry is planning to purchase a sports goods store. He calculated that in order to make profit, the average daily sales must be $> \$525$. He randomly sampled 36 days and found $\bar{X} = \$565$ and $S = \$150$

- (a) In order to estimate the average daily sales to within \$20 with 95% reliability, how many days should Jerry sample?
- (b) If the true average daily sales is \$530, what is the power of Jerry's test at the significance level of 0.05?
- (c) Suppose $\mu = \$530$. In order to guarantee $\alpha = 0.05$ and $\beta = 0.2$, how many days should Jerry sample?

Example 2: The CEO of Cereal's Unlimited Inc wants to be very certain that the mean weight μ of packages satisfies the package label weight of 16 ounces. The packages are filled by a machine that is set to fill each package to a specified weight. However, the machine has random variability measured by σ^2 . The CEO would like to have strong evidence that the mean package weight is greater than 16 oz. The quality control manager advises him to examine a random sample of 25 packages of cereal. From his past experience, the manager knew that the weight of the packages follows a normal distribution with standard deviation 0.4 oz. At the significance level $\alpha = 0.05$,

- (a) What is the decision rule (rejection region) in terms of the sample mean $\bar{X}$?
- (b) What is the power of the test when $\mu = 16.13$ oz?
- (c) How many packages of cereal should be sampled if we wish to achieve a power of 80% when $\mu = 16.13$ oz?

Example 3: The seven scores listed below are axial loads (in pounds) for a random sample of 7 12-oz aluminum cans manufactured by ALUMCO. An axial load of a can is the maximum weight supported by its sides, and it must be greater than 165 pounds, because that is the maximum pressure applied when the top lid is pressed into place.

270, 273, 258, 204, 254, 228, 282

- (a) As the quality control manager, please test the claim of the engineering supervisor that the average axial load is greater than 165 pounds. Use $\alpha = 0.05$. What assumptions are needed for your test?
- (b) Write a SAS and R program to do part (a).

Equivalence of CI and hypothesis testing

- ▶ Two-sided alternative $H_a : \mu \neq \mu_0$ at α
 $(\bar{X} - t_{n-1,\alpha/2}s/\sqrt{n}, \bar{X} + t_{n-1,\alpha/2}s/\sqrt{n})$
- ▶ One-sided alternative $H_a : \mu > \mu_0$ at α
 $(\bar{X} - t_{n-1,\alpha}s/\sqrt{n}, \infty)$
- ▶ One-sided alternative $H_a : \mu < \mu_0$ at α
 $(-\infty, \bar{X} + t_{n-1,\alpha}s/\sqrt{n})$



Inference for one population variance σ^2

Inference for one population variance σ^2 for normal distribution

- ▶ Let $X_1, X_2, \dots, X_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma^2)$
- ▶ Let $W = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ be the pivotal quantity for the inference on σ^2



Inference for one population variance σ^2 for normal distribution

Confidence interval for σ^2

$$P(\chi_{n-1, \alpha/2, L}^2 \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi_{n-1, \alpha/2, U}^2) = 1 - \alpha$$

Inference for one population variance σ^2 for normal distribution

Hypothesis test on σ^2 (one-tailed)

$$\begin{cases} H_0 : \sigma^2 \leq \sigma_0^2 \\ H_a : \sigma^2 > \sigma_0^2 \end{cases}$$



$$E(S^2) = \sigma^2$$

▶ Test statistic: $W_0 = \frac{(n-1)S^2}{\sigma_0^2} \overset{H_0}{\sim} \chi_{n-1}^2$

▶ At the significance level α , we reject H_0 if $W_0 \geq \chi_{n-1, \alpha, U}^2$

▶ p-value = $p_U = P(W_0 \geq w_0)$

Inference for one population variance σ^2 for normal distribution

Hypothesis test on σ^2 (one-tailed)

$$\begin{cases} H_0 : \sigma^2 = \sigma_0^2 \\ H_a : \sigma^2 < \sigma_0^2 \end{cases}$$

- ▶ At the significance level α , we reject H_0 if $W_0 \leq \chi_{n-1, \alpha, L}^2$ or
- ▶ p-value = $p_L = P(W_0 \leq w_0)$

Inference for one population variance σ^2 for normal distribution

Hypothesis test on σ^2 (two-tailed)

$$\begin{cases} H_0 : \sigma^2 = \sigma_0^2 \\ H_a : \sigma^2 \neq \sigma_0^2 \end{cases}$$

- ▶ At the significance level α , we reject H_0 if $W_0 \leq \chi_{n-1, \alpha/2, L}^2$ or $W_0 \geq \chi_{n-1, \alpha/2, U}^2$
- ▶ p-value = $2 \min(p_U, p_L = 1 - p_U)$

What if the population is NOT normal?

- ▶ If the population distribution is known, one can carry out the **LR test** (likelihood ratio test)
- ▶ If the population distribution is unknown, one can try Box-Cox normal transformation, or apply non-parametric procedures or resampling method (e.g. Bootstrap resampling).